

SELVA KARUPPASAMY M

AI Backend Engineer

✉ selvakaruppasamym@gmail.com ☎ 6379638115 📍 Bengaluru, Karnataka 🌐 selva-karuppasamym
🔗 SelvaKaruppasamyM

ABOUT ME

Backend Engineer with over 4 years of experience specializing in Python Tooling and high-performance System Architecture at Zentron Labs. Proven track record of bridging product vision with technical execution through cross-functional collaboration, delivering scalable, enterprise-grade solutions. Skilled in modern AI-driven development methodologies (TDD/SDD) and focused on driving measurable business value through optimized backend infrastructure.

WORK EXPERIENCE

Associate Software Developer

07/2022 – Present

Zentron Labs Pvt Ltd

- Collaborated with Product Owners to define requirements and delivered interactive Figma prototypes, cutting initial feature deployment time by ~1 week.
- Architected a Python FastAPI backend for a CVaaS platform, supporting 50 concurrent tasks and processing 100+ MB of image data.
- Modernized legacy compute functions with CFFI-based Python modules, halving image-processing latency and raising throughput to 15 fps.

SKILLS

Languages & Frameworks

Python (FastAPI, Flask, SQLAlchemy), SQLite, PostgreSQL, CFFI.

AI & Automation

OpenCode Agent, Playwright MCP, TDD, SDD.

System & Design

System Design, Microservices, Figma, REST API, Git, SVN, CI/CD Pipelines, WSL.

Vision & Device control

LabVIEW, NI-VISA, NI-IMAQdx.

PROJECTS

Agentic LabVIEW to Python Compiler

Architected a multi-stage LLM pipeline to automatically translate visual LabVIEW code into Python/C++. Built a decoupled architecture featuring a CustomTkinter GUI frontend and specialized backend agents for parsing XML logic, processing JSON data flows, and analyzing block diagram images using the Gemini API.

Real-Time Part Configurator

Developed a responsive, real-time configurator platform from the ground up using Elixir and Phoenix LiveView. Applied SDD/TDD methodologies alongside AI backend tooling to build seamless, interactive user experiences with minimal latency. Integrated Playwright MCP to facilitate advanced AI-driven development and testing workflows.

CVaaS Infrastructure Development

Built a high-performance Python backend featuring asynchronous processing, PostgreSQL database integration, and real-time health monitoring for computer vision tasks.

Industrial Configurator Architecture

Designed a modular framework for an industrial configurator integrating C++ DLLs, reducing new feature integration time from 1 week to 3 days while minimizing technical debt.

EDUCATION

Bachelor of Electrical and Electronics Engineering
Alagappa Chettiar Government College of Engineering and Technology 
CGPA: 8.6/10

07/2018 – 04/2022
TamilNadu, India